ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, robot languages and examples.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 A ✓ 1.1.9 D ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 E ✓ 1.3.5 J ✓ 1.3.6 B ✓ 1.3.7 F ✓ 1.3.8 D ✓ 1.3.9 A ✓ 1.3.10 H ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FOUR named ✓ and explained ✓: world frame – fixed reference for the whole cell; base frame – fixed at the robot's base, joint positions are measured from it; tool frame (TCP) – at the tip of the tool, moves with the tool; user / work-object frame – set on a fixture or table so points are taught relative to the work; joint coordinates – each axis angle separately. (8)

2.2 Forward kinematics: the joint angles are known and the position and orientation of the tool are calculated ✓✓. Inverse kinematics: the wanted tool position is known and the joint angles needed to reach it are calculated (may have several or no solutions) ✓✓. (4)

2.3 $\text{Arm is straight, so length} = 0{,}4 + 0{,}3 = 0{,}7\ \text{m}$ ✓; $\text{X} = 0{,}7\cos 30° \checkmark = 0{,}61\ \text{m}$ ✓; $\text{Y} = 0{,}7\sin 30° = 0{,}35\ \text{m}$ ✓ (4)

[16]

QUESTION 3

3.1 Joint move – each axis moves to its target, the path is not straight, fastest ✓ – free moves through open space ✓. Linear move – TCP moves in a straight line ✓ – approaching / leaving a part, welding a straight seam, inserting ✓. Circular move – TCP follows an arc through three points ✓ – welding or gluing round edges ✓. (6)

3.2 Logical order ✓✓✓✓✓✓, e.g.: joint move to home; joint move to the approach point above the part; linear move down to the pick point; close the gripper and wait / check sensor; linear move up to the approach point; joint move to the approach point above the place position; linear move down; open the gripper; linear move up; return home. (6)

3.3 Cycle time $= 1{,}2 + 2{,}5 + 1{,}0 + 2{,}3 = 7$ s ✓✓; parts per hour $= 3\,600 \div 7$ ✓ $= 514$ parts ✓ (4)

3.4 The robot moves quickly to a safe point above the part and then slowly and straight down ✓, so it does not hit the part, the fixture or other parts on the way in or out ✓. (2)

[18]

QUESTION 4

4.1 Lighting illuminates the part evenly ✓; a camera captures the image ✓; software processes it (finds edges, shapes, colours, measures) ✓; it compares the result with set limits and sends a decision or position to the PLC / robot (pass / reject, coordinates) ✓. (4)

4.2 Any THREE ✓✓✓: finding the position of parts for picking; checking labels and print; measuring dimensions; detecting defects; reading barcodes / QR codes; sorting by colour or shape; counting. (3)

4.3 It measures the forces and torques at the tool ✓, so the robot can apply a controlled force (e.g. for assembly, polishing, insertion) and stop if it hits something unexpected ✓. (2)

4.4 Sensor ✓ and reason ✓ each: (a) photoelectric sensor – detects any object breaking or reflecting the beam, works on cardboard; (b) vision camera / colour (contrast) sensor – checks presence and position of the label; (c) load cell – changes force into an electrical signal for weighing; (d) encoder (absolute or incremental) / resolver – measures the angle accurately. (8)

4.5 Even, stable lighting gives clear, high-contrast images, so the software can find features reliably and is not fooled by shadows or reflections ✓ (1)

[18]

QUESTION 5

5.1 Feature ✓ and example ✓ each: fixed – special machines for one product, very high volume, hard to change – bottling line, car-engine transfer line; programmable – equipment reprogrammed for each batch, medium volume – CNC machines, robots making different batches; flexible – changes products quickly with little or no downtime, mixed products – flexible manufacturing cell with robots and AGVs. (6)

5.2 Total $= 2{,}5 + 6 = 8{,}5$ kg ✓, which is less than 10 kg, so the robot is suitable (with a small safety margin) ✓ (2)

5.3 Any FIVE ✓✓✓✓✓: fixed guards / fences; interlocked doors; light curtains or laser scanners; emergency stops; safety PLC / safety relays; reduced speed in teach mode with an enabling switch; marked work envelope and warning signs; risk assessment and compliance with ISO 10218; training and lock-out procedures. (5)

5.4 Speed-and-separation monitoring: scanners measure how close a person is; the robot slows down as the person approaches and stops if they get too close ✓✓. Power-and-force limiting: the robot's speed, power and force are kept low and it stops when it detects contact, so a collision cannot injure a person ✓✓. (4)

5.5 Identify the hazards (and who may be harmed) ✓ (1)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE